



GROWN A BEAN — WATCH IT LIVE!

Detailed Lesson Plan for Young Plant Scientists

Subject: Environmental Science / Early Childhood Education

Grade Level: Kindergarten – Grade 2 (Ages 4–8) | Duration: 45–60 minutes

Curriculum Theme: Living Things, Germination & Observation Skills

LESSON AT-A-GLANCE

Subject Area	Environmental Science, Life Skills, Early Literacy
Grade / Age	Kindergarten – Grade 2 (Ages 4–8)
Duration	45–60 minutes (plus daily 5-minute observation check-ins over 7+ days)
Seed Used	Rajma (Kidney Bean) — a familiar Indian household ingredient
Key Concepts	Germination, plant needs (water / soil / sunlight), root vs. shoot, scientific observation
Materials	Rajma seeds (1 per child), disposable cups, potting soil, water, marker pens, observation journals / plain paper
Indian Context	Uses culturally familiar references: rajma-chawal, tulsi plant, mitti, morning dhoop

LEARNING OBJECTIVES

By the end of this lesson, students will be able to:

Knowledge

- Name the three things a seed needs to grow: water, soil, and sunlight.
- Explain what germination means in simple terms.

- Identify the root (goes down) and the shoot (goes up) on a seedling.

Skills

- Plant a rajma seed correctly following a 6-step procedure.
- Make daily observations and record them in a science journal.
- Draw and label simple diagrams of a growing bean plant.

Attitudes

- Show curiosity and wonder about living things.
- Develop responsibility through daily plant care.
- Connect everyday food (rajma-chawal) to the natural world.

■ CURRICULUM ALIGNMENT (NCERT / NEP 2020)

Framework	Strand / Competency	Connection to Lesson
EVS (Class 1–2)	Living & Non-living; Plants around us	Seed germination; plant parts
NEP 2020	Experiential & Activity-based Learning	Hands-on planting activity
Scientific Method	Observe → Predict → Record	Daily observation journal
Language Arts	Vocabulary development	Root, shoot, germination, seedling
Social-Emotional	Responsibility & patience	Daily plant care routine

■ MATERIALS & PREPARATION

Per Student

- 1 rajma (kidney bean) seed, soaked overnight in water
- 1 disposable plastic or paper cup (200 ml)
- Potting soil or garden mitti (~150 ml)
- Marker pen to label the cup
- Observation journal (stapled A4 sheets or exercise book)
- Pencil / crayons for drawing

Classroom Shared

- Small watering can or spray bottle
- Tray / newspaper to manage mess
- A window ledge or spot with morning sunlight
- Sample soaked seed (pre-soaked by teacher) for comparison

- Poster / slide showing root vs. shoot diagram

Teacher Prep (Day Before)

- Soak one extra rajma seed overnight to show 'waking up' effect
- Print / display slides or draw key diagrams on the board
- Prepare observation journal template (date, draw, one word)
- Test soil moisture – it should be damp, not waterlogged

■ LESSON FLOW

HOOK (5 min)

Show & Ask	Hold up a dry rajma seed. Ask: 'Have you eaten rajma-chawal? Did you know this tiny thing becomes a big plant?' Show the soaked seed side by side — children observe the size difference immediately.
Key Question	Slide 2 prompt: 'THIS tiny rajma seed... became THIS big plant! How?!'
Engagement tip	Let children touch and hold both seeds. Ask: smooth? hard? Has it changed?

DIRECT INSTRUCTION (10 min)

Three Magic Things	Introduce Water → Soil → Sunlight (Slides 5–8). Use the Indian-context comparisons: tulsi watering, mitti from the terrace, morning dhoop from the window.
Root vs Shoot	Slide 12 memory hook: 'Roots go DOWN like our feet. Shoots go UP like our hands!' Get children to stand and act it out — feet planted, hands raised.
Vocabulary wall	Write on board: seed, root, shoot, germination, seedling. Revisit each word during the activity.

HANDS-ON ACTIVITY (20 min)

Step 1	Fill cup $\frac{3}{4}$ with soil. Pat gently — 'like tucking in a blanket.'
Step 2	Poke a finger-deep hole in the soil.
Step 3	Place the rajma seed flat-side down into the hole.
Step 4	Cover with soil — 'cozy blanket.'
Step 5	Sprinkle water gently — 'soft rain, not a flood!'
Step 6	Write name on the cup. Place near the window.

OBSERVATION JOURNAL (10 min)

Day 1 Entry	Children draw their cup with the seed inside, write today's date, and one feeling word: HAPPY / EXCITED / CURIOUS.
Model it	Teacher draws on the board first to reduce anxiety about 'getting it right.'

Scaffold

Provide sentence frames for early writers: 'Today I planted _____. I feel _____.'

WRAP-UP (5 min)**Quick review**

Ask: What does our seed need? (children chorus: Water! Soil! Sunlight!)

Preview

Show the 5-stage timeline (Slide 11). 'By Day 7 you might see a green shoot!'

Take-home

Remind children to check on their plant every day — they are Plant Scientists now!

■ 7-DAY OBSERVATION TIMELINE

Children check their plant every day and record one observation. Use the table below as the journal template — photocopy one per student.

Day	What to look for	Science concept	Journal prompt
Day 1	Soil looks damp; no change visible yet	Patience & care	Draw your cup. How do you feel?
Day 2–3	Seed swells and gets noticeably bigger	Imbibition — seed absorbs water	Is your seed bigger? Draw it!
Day 3–5	Tiny white root tip appears at the bottom	Radicle emergence — first root	Can you see a root? Colour it white.
Day 5–7	Green shoot pushes upward through soil	Hypocotyl / plumule — shoot growth	A shoot appeared! Draw the green part.
Day 7+	Two seed leaves (cotyledons) unfurl	Seedling stage — leaves for photosynthesis	Count the leaves. What colour are they?
Day 10+	True leaves appear; plant grows taller	Vegetative growth begins	How tall is your plant? Use your finger!
Day 14+	Stem thickens; more leaves visible	Established seedling	Write 3 words to describe your plant.

■ DIFFERENTIATION STRATEGIES

Learner Group	Strategy	Example
Early Finishers / Advanced	Extension tasks	Compare soaked vs. dry seed weight; try growing in dark vs. light
Struggling Learners	Peer pairing & visual cues	Pair with a buddy; use picture-only journal template
ELL / Hindi-medium	Bilingual vocabulary	Post word wall in English + Hindi (■■■■ = root, ■■■■■■ = shoot)
Special Needs	Sensory-friendly adaptation	Pre-fill cup with soil; use large-grip pencil for journal

■ ASSESSMENT & EVALUATION

Formative (Ongoing)

- Daily journal check: Has the child drawn and dated their entry?
- Oral questioning during lesson: 'What does your plant drink? Where does its food come from?'
- Thumbs up / thumbs down quick checks at key vocabulary points.

Summative (End of Week 2)

- Completed observation journal (7+ entries) — look for progression in drawings.
- Label a blank plant diagram: root, shoot, leaf, soil.
- One-sentence oral explanation: 'My plant grew because...'

Success Criteria

- Can name 3 things a plant needs (water, soil, sunlight).
- Can point to and name root and shoot on their seedling.
- Has at least 5 journal entries with a drawing and date.

■ TEACHER TIPS & COMMON ISSUES

■ Seed doesn't sprout by Day 5

Check: Is the soil damp (not dry or waterlogged)? Is it getting at least 2 hours of light? Try gently removing the seed — if it smells bad, it may have rotted. Start fresh.

■ Children overwater

Demonstrate the 'sprinkle test' — 5 sprinkles from a watering can. Make a class rule: 'We water ONLY if the soil feels dry when you touch it.'

■ Journal reluctance

*Emphasise that scientist drawings don't have to be 'perfect.' Show famous scientist sketchbooks.
Reward effort, not accuracy.*

■ Indian summer heat

In hot climates (Madurai etc.), keep the plant away from direct afternoon sun after Day 3. Morning light (before 10 am) is ideal; harsh afternoon sun may dry the soil too quickly.

■ Home extension

Send a simple Parent Letter (see below) inviting families to photograph the plant each day and share images with the class — builds community and responsibility.

■ KEY VOCABULARY

Term	Definition
Seed / Beej (■■■■)	A tiny structure containing a baby plant; the starting point of plant life.
Germination / ■■■■■■	The process by which a seed begins to sprout and grow into a new plant.
Root / Jad (■■■)	The part of the plant that grows downward into the soil to absorb water and nutrients.
Shoot / Ankur (■■■■■)	The part that grows upward toward the light; includes the stem and leaves.
Seedling / Paudha	A very young plant that has just germinated and begun to grow.
Soil / Mitti (■■■■■■)	The layer of earth in which plants grow; provides anchor and nutrients.
Sunlight / Dhoop (■■■)	Light energy from the sun; used by leaves to make food through photosynthesis.
Observation / Nirikhshan	Careful watching and noting of changes over time — the core skill of science.

■ PARENT / GUARDIAN COMMUNICATION LETTER

Dear Parent / Guardian,

Your child has planted a rajma (kidney bean) seed today as part of our science lesson 'Grown a Bean — Watch It Live!' They are learning about how seeds germinate and what plants need to grow: water, soil, and sunlight.

What you can do at home:

- Ask your child to check the plant every morning and tell you what they see.
- Help them water it gently — just a little sprinkle — if the soil feels dry.
- Place the cup on a windowsill that gets morning light.
- Take a photo each day and share it with us — we'd love to see it grow!

This activity builds observation skills, scientific curiosity, and a sense of responsibility. The whole family can join in the wonder of watching a tiny seed wake up!

Thank you for your support,

Class Teacher

■ OBSERVATION JOURNAL TEMPLATE (Photocopy for students)

Use one row per day. Encourage drawing in the box and writing the date.

Day	Date	Draw what you see	One word: How does it look?	Did you water today?
1				■ Yes ■ No
2				■ Yes ■ No
3				■ Yes ■ No

4				■ Yes ■ No
5				■ Yes ■ No
6				■ Yes ■ No
7				■ Yes ■ No
8				■ Yes ■ No
9				■ Yes ■ No
10				■ Yes ■ No

*Lesson plan based on the 'Grown a Bean — Watch It Live!' slide deck. Designed for Indian early-childhood classrooms.
Freely adaptable for classroom use.*